Light

GCSE Computer Science — Memory and Storage (Set 3)

Over **1,100,000+ marks** of exam-aligned content across **67 qualifications**. 4,200+ auto-marked assessments & long-answer practice. No login required.

GCSE • Edexcel • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

00:00 Start Pause Reset

← Back to Memory And Storage

Multiple-choice topic test for **GCSE Computer Science (Edexcel)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

A laptop computer has 16 GiB of RAM and a 512 GiB SSD.
Why does the laptop need both RAM and secondary storage?

- ○ **A.** RAM stores files permanently while the SSD stores temporary data
- ○ **B.** RAM is used for long-term storage while the SSD runs programs
- ○ **C.** RAM holds data temporarily while programs run; the SSD stores data permanently even when powered off
- ○ **D.** RAM and SSD perform identical functions but the SSD is faster

Q2 (1 mark)

A computer's RAM is described as volatile memory.
What does 'volatile' mean in this context?

- ○ **A.** The memory loses all its data when the power is turned off
- ○ **B.** The memory is dangerous and could explode if overheated
- ○ **C.** The memory can only store temporary files, not programs
- ○ **D.** The memory changes speed depending on how much data it holds

Q3 (1 mark)

A smartphone stores photos on its internal flash storage.
How does flash storage physically store data?

- ○ **A.** By magnetising tiny areas on a spinning disk
- ○ **B.** By recording vibrations onto a crystal wafer
- ○ **C.** By burning patterns of holes into a plastic surface
- ○ **D.** By trapping electrical charges in microscopic memory cells

Q4 (1 mark)

A traditional hard disk drive stores data magnetically.
What physical components are inside an HDD?

- ○ **A.** Memory chips and a USB controller
- ○ **B.** Spinning platters and a moving read/write head
- ○ **C.** A laser and a reflective disc
- ○ **D.** Flash memory cells and a circuit board only

Q5 (1 mark)

A DVD player reads data from a DVD disc.
What technology does the DVD player use to read the data?

- ○ **A.** A magnetic sensor that detects magnetised areas
- ○ **B.** An electrical probe that measures charge in cells
- ○ **C.** A mechanical needle that follows grooves in the disc
- ○ **D.** A laser beam that detects reflections from the disc surface

Q6 (1 mark)

Data storage uses various units of measurement.
What is a nibble?

- ○ **A.** A group of 4 bits
- ○ **B.** A group of 2 bits
- ○ **C.** A group of 8 bits
- ○ **D.** A group of 16 bits

Q7 (1 mark)

Storage capacity is measured using binary multiples.
How many bytes are in one kibibyte (KiB)?

- ○ **A.** 100 bytes
- ○ **B.** 1000 bytes
- ○ **C.** 1024 bytes

- ○ **D.** 1048 bytes

Q8 (1 mark)

A student wants to email several large photos to a friend, but the total file size exceeds the email attachment limit.
What technique should the student use to reduce the total file size?

- ○ **A.** Compression
- ○ **B.** Defragmentation
- ○ **C.** Encryption
- ○ **D.** Formatting

Q9 (1 mark)

JPEG is a common image format that uses lossy compression.
What happens to an image when it is saved as a JPEG?

- ○ **A.** The image is encrypted so only authorised users can view it
- ○ **B.** The image doubles in size to improve quality
- ○ **C.** The image is converted to black and white to save space
- ○ **D.** Some image data is permanently removed to reduce file size

Q10 (1 mark)

ZIP is a common file format used to compress folders of files.
What type of compression does ZIP use?

- ○ **A.** Lossy compression, which permanently removes some data
- ○ **B.** Temporary compression, which only works while files are open
- ○ **C.** Physical compression, which makes the storage device smaller
- ○ **D.** Lossless compression, which allows exact restoration of the original files

Q11 (1 mark)

A gamer upgrades their computer from an HDD to an SSD.
What improvement will they most likely notice?

- ○ **A.** Games will have better graphics because SSDs process images faster
- ○ **B.** Games and the operating system will load much faster
- ○ **C.** The computer will use less electricity because SSDs need more power
- ○ **D.** More games can be stored because SSDs have larger capacity

Q12 (1 mark)

A charity distributes free educational software on CDs.
Why is optical storage suitable for this purpose?

- ○ **A.** CDs have the largest storage capacity of any medium
- ○ **B.** CDs are inexpensive to produce in large quantities and are portable
- ○ **C.** CDs can be updated remotely after distribution

- ○ **D.** CDs are faster than all other storage types

Q13 (2 marks)

A video file is 3072 MiB in size.
How many gibibytes (GiB) is this?

- ○ **A.** 3.072 GiB, because 3072 ÷ 1000 = 3.072
- ○ **B.** 0.003 GiB, because MiB is much smaller than GiB
- ○ **C.** 3,145,728 GiB, because 3072 × 1024 = 3,145,728
- ○ **D.** 3 GiB, because 3072 ÷ 1024 = 3

Q14 (2 marks)

A memory card has a capacity of 32 GiB.
How many mebibytes (MiB) is this?

- ○ **A.** 32,000 MiB, because 32 × 1000 = 32,000
- ○ **B.** 32 MiB, because GiB and MiB are equal
- ○ **C.** 0.03125 MiB, because 32 ÷ 1024 = 0.03125
- ○ **D.** 32,768 MiB, because 32 × 1024 = 32,768

Q15 (2 marks)

A laptop user frequently travels with their computer.
Why might an HDD be unsuitable for a portable laptop?

- ○ **A.** HDDs cannot be connected to laptops due to size constraints

- ○ **B.** HDDs have moving parts that can be damaged by movement and vibration
- ○ **C.** HDDs require constant internet connection to function
- ○ **D.** HDDs can only store data when the laptop is plugged into mains power

Q16 (2 marks)

A data centre manager is considering replacing all HDDs with SSDs. What is the main disadvantage of this approach?

- ○ **A.** SSDs are slower than HDDs for all operations
- ○ **B.** SSDs cannot store as much data as HDDs in the same physical space
- ○ **C.** SSDs cost significantly more per gigabyte than HDDs
- ○ **D.** SSDs require special cooling systems that HDDs do not need

Q17 (2 marks)

A user needs to choose between CD, DVD, and Blu-ray for backing up 20 GiB of photos.
Which is the most practical choice and why?

- ○ **A.** CD, because CDs are the most reliable optical format
- ○ **B.** DVD, because a single DVD can hold 20 GiB of data
- ○ **C.** Any format works because all optical discs have the same capacity

○ **D.** Blu-ray, because a single Blu-ray disc can hold 25 GiB, fitting all photos on one disc

Q18 (2 marks)

A web developer is optimising images for a website.
Why would they choose lossy compression for photographs but lossless compression for company logos?

○ **A.** Photographs cannot use lossless compression due to technical limitations

○ **B.** Logos require faster loading times than photographs

○ **C.** Lossy compression only works on photographs; lossless only works on graphics

○ **D.** Photographs tolerate quality loss without noticeable impact; logos need sharp edges and exact colours

Q19 (2 marks)

A musician wants to burn demo tracks onto discs. They might need to replace some tracks later if they record better versions.
Which optical disc format should they use?

○ **A.** CD-ROM, because it offers the best audio quality

○ **B.** CD-R, because it can be written once and is widely compatible

○ **C.** CD-RW, because tracks can be erased and replaced with new recordings

○ **D.** DVD-ROM, because it holds more music than CDs

Q20 (2 marks)

A podcast consists of 5 episodes. Each episode is 64 MiB.
What is the total size of the entire podcast?

- ○ **A.** 69 MiB, because 64 + 5 = 69
- ○ **B.** 320 MiB, because 64 × 5 = 320
- ○ **C.** 12.8 MiB, because 64 ÷ 5 = 12.8
- ○ **D.** 327,680 KiB, because 64 × 5 = 320 MiB = 320,000 KiB

Q21 (2 marks)

A computer system uses three types of secondary storage: magnetic, optical, and solid state.
Which list correctly matches storage devices to their types?

- ○ **A.** Magnetic: SSD; Optical: HDD; Solid state: DVD
- ○ **B.** Magnetic: DVD; Optical: USB flash drive; Solid state: HDD
- ○ **C.** Magnetic: HDD; Optical: Blu-ray; Solid state: USB flash drive
- ○ **D.** Magnetic: CD; Optical: SSD; Solid state: HDD

Q22 (2 marks)

A file was 500 MiB before compression. After compression, it is 400 MiB.
What percentage of the original file size has been saved?

- ○ **A.** 80%, because 400 ÷ 500 = 0.8
- ○ **B.** 20%, because (500 - 400) ÷ 500 = 0.2

- ○ **C.** 25%, because 100 ÷ 400 = 0.25
- ○ **D.** 125%, because 500 ÷ 400 = 1.25

Q23 (2 marks)

An SSD manufacturer states their drive is rated for 600 TBW (terabytes written).
What does this rating indicate?

- ○ **A.** The drive can store up to 600 terabytes of data at once
- ○ **B.** The drive will fail exactly when 600 terabytes have been written to it
- ○ **C.** The drive can reliably handle approximately 600 terabytes of total write operations over its lifespan
- ○ **D.** The drive transfers data at 600 terabytes per second

Q24 (3 marks)

A scanner produces images with these settings:
• Resolution: 2400 × 1600 pixels
• Colour depth: 32 bits per pixel
What is the uncompressed file size of one scanned image?

- ○ **A.** 15,360,000 bytes, because 2400 × 1600 × 4 = 15,360,000
- ○ **B.** 3,840,000 bytes, because 2400 × 1600 = 3,840,000 pixels
- ○ **C.** 122,880,000 bytes, because 2400 × 1600 × 32 = 122,880,000
- ○ **D.** 1,920,000 bytes, because 2400 × 1600 ÷ 2 = 1,920,000

Q25 (3 marks)

A file server has 16 TiB of total storage capacity.
How many gibibytes (GiB) is this?

- ○ **A.** 16,000 GiB, because 16 × 1000 = 16,000
- ○ **B.** 16,384 GiB, because 16 × 1024 = 16,384
- ○ **C.** 0.015625 GiB, because 16 ÷ 1024 = 0.015625
- ○ **D.** 16,777,216 GiB, because 16 × 1024 × 1024 = 16,777,216

Q26 (3 marks)

A voice recording has these properties:
• Sample rate: 22,050 samples per second
• Bit depth: 8 bits per sample
• Duration: 5 minutes
• Mono (1 channel)
What is the uncompressed file size in bytes?

- ○ **A.** 6,615,000 bytes, because 22,050 × 8 × 300 × 1 ÷ 8 = 6,615,000
- ○ **B.** 52,920,000 bytes, because 22,050 × 8 × 300 = 52,920,000 (without bit conversion)
- ○ **C.** 882,000 bytes, because 22,050 × 8 × 5 ÷ 8 = 882,000 (using minutes not seconds)
- ○ **D.** 22,050 bytes, because sample rate determines file size directly

Q27 (3 marks)

A folder of documents is 120 MiB uncompressed. After lossless compression, it is 30 MiB.
What is the compression ratio and the space saved?

- ○ **A.** Compression ratio 4:1; space saved 90 MiB (75%)
- ○ **B.** Compression ratio 1:4; space saved 30 MiB (25%)
- ○ **C.** Compression ratio 30:120; space saved 90 MiB (75%)
- ○ **D.** Compression ratio 3:1; space saved 60 MiB (50%)

Q28 (3 marks)

A security company needs storage for CCTV footage with these requirements:
• Continuous recording 24/7
• Must retain footage for 30 days
• Daily footage is approximately 50 GiB
• Cost is the primary concern
Which storage solution is most appropriate?

- ○ **A.** A 1 TiB SSD, because SSDs are the most reliable for continuous writing
- ○ **B.** Multiple 2 TiB HDDs, because they provide sufficient capacity at the lowest cost per gigabyte
- ○ **C.** A stack of DVD-RW discs, because they can be rewritten and reused indefinitely
- ○ **D.** A 512 GiB USB flash drive, because it is portable and easy to replace

Q29 (3 marks)

A photographer has 450 GiB of photos. Their external drive has 460,800 MiB of free space.
Will all the photos fit on the drive?

- ○ **A.** Yes, because 460,800 is a larger number than 450, so there is

plenty of space

- ○ **B.** No, because 460,800 MiB is less than 450 GiB when converted
- ○ **C.** Yes, because 460,800 MiB = 450 GiB, matching the requirement exactly
- ○ **D.** No, because MiB cannot be directly compared to GiB

Q30 (3 marks)

A media company needs storage for three purposes:
• Purpose A: Video editing workstation (fast random access essential)
• Purpose B: Completed project archive (large capacity, rarely accessed)
• Purpose C: Sending review copies to clients
Which combination of storage is most suitable?

- ○ **A.** A: HDD; B: SSD; C: CD-ROM
- ○ **B.** A: DVD-RW; B: USB flash drives; C: HDD
- ○ **C.** A: SSD; B: HDD; C: USB flash drive or cloud link
- ○ **D.** A: Blu-ray; B: CD-R; C: SSD

Total: **55** marks • Time: **30 min**

↑ Back to top

Submit answers

Williams Physics Education — over **1,100,000+ marks** of exam-aligned content and **4,200+ auto-marked assessments** across 67 qualifications at GCSE, A-Level, AP & Professional Qualifications. Free to use, no login required.

About & Contact • Privacy • Terms • Accessibility • Copyright • DMCA • YouTube

WPE is free for everyone. If our resources help you or your students, you can support the project here.

© 2026 Williams Physics Education. All rights reserved.